Motion

A campus activity app that helps Obafemi Awolowo University students find running routes and group runs, so nobody has to run alone.

The problem it solves

Running alone on a large campus can feel unsafe or unmotivating, and there is no single place to find where people run or who is going. Motion gives students a shared list of common OAU (Ile-Ife) routes and a simple way to organise and join group runs. It is designed for cheap Android phones and slow data: small pages, map code that loads only when the map scrolls into view, and backend libraries that download only when a backend is configured.

Key features (built in the code)

- **Campus route map.** Six named routes (Road One, Sports Centre loop, Motion Ground circuit, Amphitheatre steps, Hostels run, Staff quarters loop) with level, terrain and best-time details. Route distance is calculated from each path using the haversine formula.
- **Schedule a group run.** Pick a route, date and time (Lagos time), pace (Walk / Easy / Steady / Fast), group size (2–50), a meeting-point note, and an optional women-only label. Runs can be booked up to 60 days ahead.
- **Join, leave and cancel.** A roster shows who is going; private profiles appear as “A runner”. Joining is refused when a run is full, cancelled or over.
- **WhatsApp sharing** of a run, and a “My runs” profile page with name and privacy settings.
- **Sign-in.** Google and email/password (Firebase), magic-link (Supabase), or instant sign-in in demo mode.
- **Explore + waitlist.** Nine upcoming activity categories, each with a “Notify me” button that records interest.
- **Route recorder tool.** A hidden page that records a route by GPS walking or tap-to-draw and exports JSON to replace draft route paths.
- **Demo mode.** With no backend configured, the whole app works from browser storage with seeded sample runs.

Tech stack

Front end	Next.js 15 (App Router), React 19, TypeScript, plain CSS; installable web app manifest
Backend (choose one)	Firebase (Auth + Firestore, with security rules) – or Supabase (Postgres, row-level security, views, triggers)
Maps	Leaflet with free CARTO tiles by default; Google Maps JS API when a key is set
Hosting	Vercel (build settings pinned in `vercel.json`; Node 22)

How it works

Motion is a Next.js web app. Every screen talks to data through one `Store` interface (`lib/store/types.ts`), and three interchangeable implementations sit behind it: a browser-storage demo store, a Firebase store and a Supabase store. At start-up the app checks which environment settings exist and loads only the matching one, so a phone never downloads a backend it does not use.

Before anything is saved, shared checks in `lib/store/validate.ts` enforce the same rules the databases enforce (valid route and pace, 2–50 people, future date within 60 days, note length). The databases then enforce them again: Firestore security rules and Supabase row-level security ensure that only a host can create or cancel their run, people can only add or remove themselves, and private profiles never expose a name. Routes themselves are fixed data in `lib/routes.ts`, drawn on a lazily loaded map component that switches between Leaflet and Google Maps.

> **In short:** screens → one Store interface → demo, Firebase or Supabase, with the same validation and privacy rules applied at both the app and database level.

Current status

Built: the full v1 flow – browse routes, sign in, schedule, join, leave, cancel and share runs, profile and privacy, waitlist sign-ups, route recorder, three backends, database rules/schema, and Vercel deployment configuration.

In progress / not finished:

- All six route paths are approximate drafts (`verified: false`) and are labelled “Draft route” in the app until they are recorded on the ground with the route recorder.
- The nine Explore categories are waitlist placeholders only; none of those activities is built.
- No automated tests exist in the repository (only a type-check script).
- No live deployment, user numbers or usage data are recorded in the repository, so none are claimed here.

Next steps

- Walk and record the real route paths, then mark them verified.
- Add run reminders and run chat (listed as planned in the project README).
- Set up a production backend (Firebase project and published rules) and a custom email sender if Supabase is used.
- Add automated tests for validation and the store implementations.
- Build the highest-demand Explore category, using waitlist counts to decide.

> **Prepared by Amazoe Solutions**, business systems and software consultancy, Osogbo, Osun State. Content in this overview is drawn from the project's source code and README.